

CHAPTER ONE

AUTOMATIC WHEELCHAIR - AN OVERVIEW

1.1 INTRODUCTION

These days, thanks to embedded systems and cheap microcontrollers, automating assistive devices is easier than ever. Regular wheelchairs still need you to push them yourself, which isn't easy for elderly folks or people with disabilities. They end up having to rely on caregivers, which can knock their confidence and comfort. That's where the Automatic Wheelchair steps in. Built around the ESP32 microcontroller, this wheelchair accepts joystick commands and supports GPS navigation. The user stays in control, but the system steps in to make things simpler, safer, and more automatic. It's got more than just movement covered. There are IR sensors to spot obstacles, emergency alerts, and GPS location sharing. You get safety and peace of mind, but you're still the one making decisions. The chair won't just roll off on its own.

1.2 WHAT IS AN AUTOMATIC WHEELCHAIR?

An automatic wheelchair takes a lot of the work out of getting around. Instead of pushing the wheels by hand or relying on someone else, you use a joystick or even GPS directions. The electronics inside pick up on what you want and handle the rest.

It's not about the chair moving completely on its own. The real goal is to make movement easier, especially for people who struggle with strength or mobility. Inside, you've got sensors, microcontrollers, and motors working together. You give the command, and the chair adjusts itself, simple as that. It means folks who have trouble using a regular wheelchair can still get around on their own, without feeling worn out or unsafe.

The control system keeps things smooth, steers around obstacles, and even helps out in emergencies. With extras like ESP32 chips, ultrasonic sensors, and GPS or GSM modules, the chair does more than just move. It can communicate, navigate, and support people in real time. That makes it a solid choice for seniors, people with disabilities, or anyone in a hospital or rehab centre who needs a little extra help.

1.3 AUTOMATIC WHEELCHAIR

The Automatic Wheelchair makes life a lot easier for older adults and people with disabilities. It's all about giving users the freedom to move around on their own, without a lot of heavy lifting or help from others. You can control it in a bunch of ways, joystick or even GPS—so there's something for everyone. Just pick what works best, and the wheelchair handles the rest, moving where you want it to go.

At the heart of it all, the ESP32 microcontroller keeps things running smoothly. It collects signals from the different controls and instantly turns them into movement. You don't have to wait around; the response is quick and feels natural. Plus, the wheelchair always keeps an eye out for obstacles. Thanks to ultrasonic sensors, it constantly checks what's around you and stops if something gets in the way, so you're less likely to bump into anything.

Security's covered too. The system comes with built-in GSM and GPS features. If something's wrong—maybe there's unusual movement, or someone hits the emergency button—it sends an alert straight to caregivers. The alert goes out as a text or WhatsApp message and includes your exact location, so help can get to you fast.

You also get clear feedback from the wheelchair itself. LEDs and a buzzer let you know what's going on—like if there's an obstacle, if the system needs attention, or in case of emergencies. There's a screen onboard too, so you can always check the control mode, battery status, or just see how things are running.

Altogether, this wheelchair isn't just about getting from point A to point B. It's built for safety, easy communication, and peace of mind—for both users and their families. It works well in homes, hospitals, rehab centres, and elder-care facilities, and it doesn't break the bank either.

1.4 BENEFITS OF AUTOMATIC WHEELCHAIR CONTROL

Let's talk about what happens when you swap out ultrasonic sensors for IR sensors in an automatic wheelchair. You're still getting a bunch of big benefits: comfort, safety, and more independence for people who can't always get around on their own. The wheelchair takes a lot of the physical work out of the equation, relying on smart tech to help users move with ease. Here's how it shakes out:

1. Effortless Operation

You don't have to push the wheelchair yourself anymore. Just move a joystick, speak a simple command, or use a gesture—whatever works best—and the chair responds. No more worrying about getting tired out from just trying to get around.

2. Accessible for Different Abilities

Since there's more than one way to control the chair, people with all sorts of challenges, trouble moving their hands, issues with speech, or even tremors, can still operate it without a hassle.

3. Safety with IR Sensors

Here's where the IR sensors come in. The wheelchair constantly sends out tiny beams of infrared light. When these bounce back from something nearby, a wall, a chair, a person, the chair knows it's there and can steer clear. This kind of sensing is especially handy indoors, helping to avoid bumps and crashes in tight spaces.

4. Emergency Alerts

If something goes wrong—maybe a fall or a health scare—the built-in GSM and GPS kick in. The chair automatically sends a message with the user's location to their caregiver. That way, help isn't far behind.

5. More Independence

With all these features, users don't have to wait for someone else to help them move. They get to go where they want, on their own terms, which does wonders for confidence and independence.

6. Real-Time Feedback

LEDs, buzzers, and a display screen keep the user in the loop. You'll know if there's an obstacle, if the battery's running low, or if there's something else you need to pay attention to. No surprises.

7. Low Power, Affordable

IR sensors, DC geared motors, and ESP32 chips don't use much energy and don't cost much. So, the wheelchair stays affordable and easy to care for in the long run.

1.5 APPLICATIONS OF AUTOMATIC WHEELCHAIR CONTROL

You can use the Automatic Smart Health-Assisted Wheelchair in all sorts of real-life situations where people need help getting around, staying safe, and keeping things simple. Thanks to its multi-control setup and those IR sensors that spot obstacles, it fits right into both personal and medical settings. Here's where it really shines:

- **Hospitals and Medical Facilities**

Doctors and nurses use it to move patients who can't walk or have limited movement. It takes the heavy lifting off attendants and lets patients get around more easily.

- **Therapy Centres**

People working on their recovery use the wheelchair to practice moving in a safe, controlled way. It builds confidence and speeds up progress.

- **Homes and Elder-Care Centres**

For older folks who struggle with manual wheelchairs, this chair means they can move around the house on their own, no pushing required.

- **Individuals with Permanent Disabilities**

Anyone with spinal injuries, muscular dystrophy, or limb disabilities can run the wheelchair themselves—using a joystick, their voice, or even gestures.

- **Indoor Navigation in Smart Homes**

Because of the automatic controls and IR sensors, it glides through tight spots, dodges furniture, and handles tricky corners inside the house.

- **Emergency and Assisted Living Situations**

With GPS and a GSM alert system built in, caregivers can keep an eye on users and jump into action fast if there's an emergency.

1.6 HOW THE AUTOMATIC WHEELCHAIR WORKS

The Automatic Smart Health-Assisted Wheelchair puts the user in control, but it's always looking out for safety. You pick where you want to go—forward, back, left, or right—using a joystick, your voice, hand gestures, or even GPS directions. Once you've made a choice, the

ESP32 microcontroller jumps into action. It grabs your input, figures out what you want, and tells the motors what to do.

Here's how the system works, step by step:

- **User Input Command**

You get options. Joystick? Wi-Fi? Just pick a mode, and the wheelchair listens. The ESP32 grabs your command and gets ready to roll.

- **Signal Processing**

The ESP32 doesn't just take instructions—it thinks. It turns your direction into signals for the motor driver while keeping an eye on all the system's sensors.

- **IR Sensor-Based Obstacle Detection**

IR sensors are always on. They shoot out invisible rays, and if something bounces those rays back, the sensor spots it and tells the ESP32. If there's something in your way, the chair slows down or stops, and you'll see a light or hear a buzzer. No surprises—just safe travels.

- **Motor Control with L298N Driver**

Once the path is clear, the ESP32 sends the right signals to the motor driver. This controls the two DC geared motors, deciding how fast you move, which way you turn, and when you stop. Everything feels smooth and stable.

- **Emergency Alerts**

If something goes wrong or you need help fast, the GSM module steps in. It sends an SMS or WhatsApp message, complete with your live GPS location, to your caregiver. They get the alert instantly, so help is always close.

- **Visual and Audio Feedback**

LEDs and buzzers keep you in the loop—obstacles, emergencies, power status, and control mode changes. There's also a display (LCD or OLED) giving you real-time info about how things are going.

With everything working together, you get an intelligent wheelchair that's easy to use, safe, and always connected.

1.7 NEED FOR AUTOMATIC WHEELCHAIR

Getting around on your own matters a lot. Most of us take it for granted, but for many older adults and people with disabilities, manual wheelchairs can be a real struggle. Pushing yourself everywhere gets exhausting fast, and it's not always possible. That kind of daily effort wears people down. Before long, they have to lean more on caregivers, lose some freedom, and end up pretty uncomfortable, especially if they need to sit for hours.

Automatic wheelchairs change the game. Instead of all that pushing, users just move a joystick or tap on a phone app. No more heavy work—just easy, smooth movement. It's not just about comfort, either. It's about independence.

There's another big reason these systems matter: safety indoors. Think about crowded hospital rooms or tight hallways at home. Navigating around furniture and equipment in a manual wheelchair is tough. Add in IR sensors, and now the wheelchair can spot obstacles and avoid bumps or crashes. That's a big relief for users and their families.

And let's not forget the people who help out—nurses, family, or other caregivers. Automatic wheelchairs lighten their load, which is huge when staff are stretched thin. Plus, the emergency alert system is a lifesaver. If something goes wrong—like a health scare—the wheelchair can send out a message through GSM, so help arrives fast. That kind of backup gives everyone more peace of mind.

In the end, automatic wheelchairs make life easier and safer. They cut down on physical strain, boost confidence, and let people hold on to their independence. They really are essential for anyone dealing with mobility challenges.

CHAPTER TWO

TECHNOLOGY USED

2.1 INTRODUCTION

Every embedded system tackles a specific job, blending hardware and software to get it done. The Automatic Smart Health-Assisted Wheelchair takes this idea and runs with it. It uses embedded tech to read user commands, then manages how the wheelchair moves by controlling motors, keeping tabs on sensors, and talking to other devices. For this project, the ESP32 microcontroller sits at the heart of things. It takes care of everything—joystick controls, Wi-Fi commands from a phone app, input from IR sensors, and running the motors. The system reacts instantly, making sure the ride stays safe and smooth for older adults or anyone with mobility challenges.

Bringing together microcontrollers, sensors, motor drivers, and communication modules turns this wheelchair into something efficient and reliable. It's easy to use, especially indoors, and really makes getting around simpler for people who need it.

2.2 BLOCK DIAGRAM

The block diagram for the circuit is shown in Fig. 2.1.

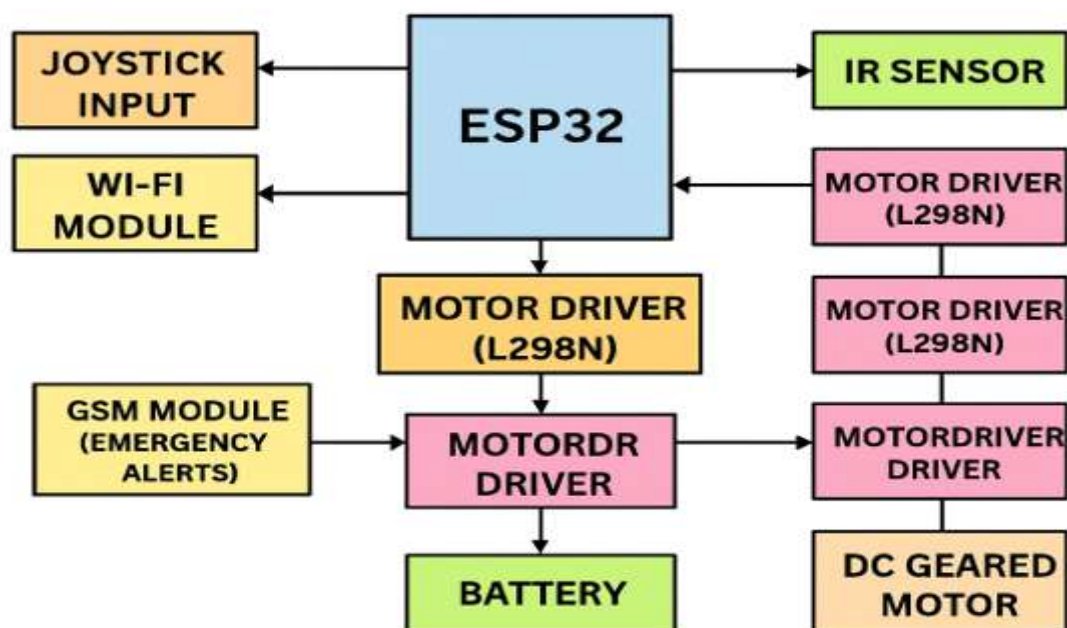


Fig. 2.1 Block Diagram

2.3 COMPONENT LIST

Here we explain the components used to design the whole system with detailed specifications. Technology used and particulars are also given regarding each of the components.

The hardware developed for the present system consists of the following main modules and circuitry:

- ESP32 microcontroller board (2 units)
- Joystick module
- Wi-Fi communication module (Inbuilt in ESP32)
- IR obstacle detection sensors
- L298N motor driver modules (4 units)
- DC geared motors (4 units)
- GSM module (for emergency alerts)
- Battery / Power supply
- LCD / OLED display
- Buzzer
- LED indicators
- Switches and push buttons
- Wires and connectors
- Wheelchair frame / chassis

2.4 ESP32 MICROCONTROLLER

The ESP32 microcontroller runs the show in the Automatic Wheelchair system. It's a dual-core powerhouse with built-in Wi-Fi and Bluetooth, so it's perfect for anything that needs wireless control and fast responses. In this setup, there are two ESP32 boards. The first one does most of the heavy lifting—taking input from the joystick, reading signals from IR sensors, controlling four motor drivers, and sending out emergency alerts through the GSM module. The second board handles the Wi-Fi side, letting users steer the wheelchair wirelessly from their phones.

The ESP32 packs a lot into a small chip: a fast processor, plenty of GPIO pins, ADC and PWM support, and it talks to other devices using protocols like UART, I2C, and SPI. It runs on 3.3V and has enough flash memory for all the control code. You can power it through a regulated supply or just use the VIN pin, so it fits easily into battery-powered projects. Programming it is simple—just use the Arduino IDE with C or C++. Tons of libraries are available for sensors, motors, and communication modules. The ESP32 keeps everything running at once: reading sensor data, controlling motors, and handling Wi-Fi, all at the same time. That’s why it’s such a great fit for the Automatic Wheelchair project.

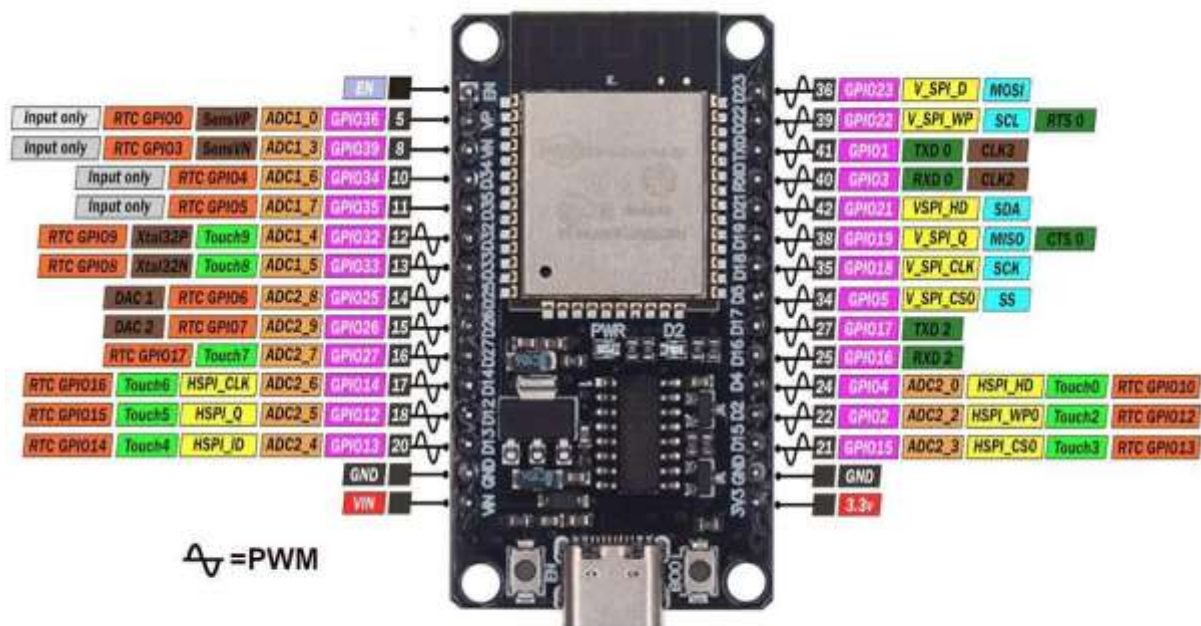


Fig. 2.2 ESP32

2.5 JOYSTICK MODULE

The joystick module gives users hands-on control over the Automatic Wheelchair. Push it—forward, back, left, or right—and it sends out analog voltage signals that shift with your movement. The ESP32 picks up these signals through its analog inputs, figures out which way you want to go, and tells the motors what to do. It feels smooth and natural in your hand, which is great for anyone who likes direct, simple control. The design is straightforward, it hooks up to the ESP32 without any fuss, and it just works—making it a key part of the wheelchair system.



Fig. 2.3 JOYSTICK MODULE

2.6 IR SENSOR

The IR sensor spots obstacles in front of the automatic wheelchair. It sends out infrared light and picks up the signal that bounces back when something's close. When it finds an obstacle, it alerts the ESP32, so the controller can stop or slow down the wheelchair and avoid a crash. IR sensors are simple and cheap, and they work well for short distances indoors. That's why they're perfect for keeping users safe at home or in hospitals.

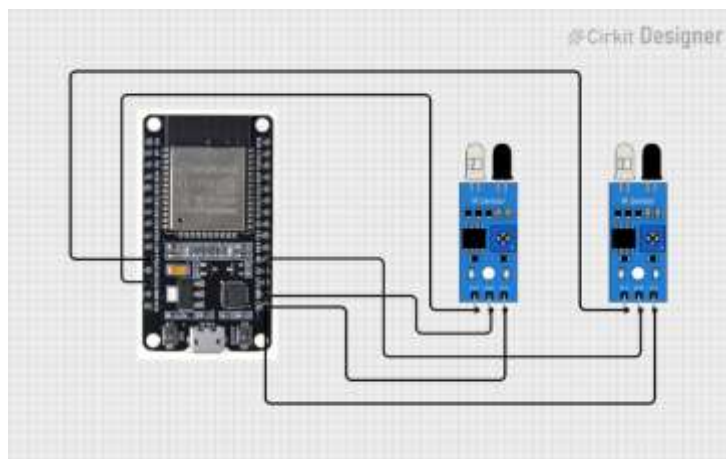


Fig. 2.4 IR SENSOR

2.7 L298N MOTOR DRIVER

The L298N motor driver handles the DC geared motors on the Automatic Wheelchair. The ESP32 can't push enough current for the motors on its own, so the L298N steps in as the

middleman. With it, the ESP32 sends simple signals and easily manages both speed and direction for each wheel. Here, there's one L298N for every motor—four in total—so the chair moves forward, backward, or turns without a hitch. The module's reliable, easy to set up, and delivers all the power those motors need to run smoothly.

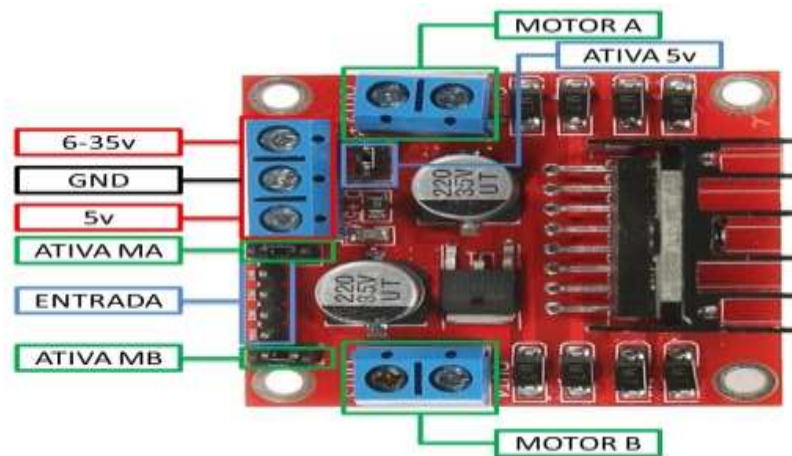


Fig. 2.5 L298N MOTOR DRIVER

2.8 DC GEARED MOTOR

The DC geared motors power the wheels on the Automatic Wheelchair. They deliver plenty of torque at low speeds, so they can handle both the weight of the user and the chair itself. Each wheel gets its own motor, which means the wheelchair moves smoothly and turns easily in any direction. The motors team up with L298N motor drivers, letting the ESP32 control forward, backward, and turning motions without a hitch. They're tough, reliable, and pack enough punch to handle mobility tasks without breaking a sweat.

2.9 GSM Module

The GSM module lets the Automatic Wheelchair send out emergency alerts straight to caregivers or family. If the user presses the emergency button or the system notices something's wrong, the ESP32 jumps into action and tells the GSM module to send a message—SMS or WhatsApp—with the user's info. So, even if the person's alone, help gets notified fast. The module uses a SIM card and talks to the ESP32 over UART. It's easy to set up and works well for real-time alerts.



Fig. 2.6 GSM MODULE

2.10 LIQUID CRYSTAL DISPLAY (LCD)

Liquid Crystal Display (LCD) screen is an electronic display module which finds a wide range of applications. 16x2 LCD display is a very basic module and is very commonly used in various devices and circuits. These modules are preferred over seven segments and other multi segment LEDs. The reasons being, LCDs are economical, easily programmable, have no limitations of displaying special and even custom characters (unlike in seven segments), animations and lot of other useful features. A 16x2 LCD means it can display 16 characters per line and there are 2 such lines. In this LCD each character is displayed in 5x7 pixel matrix. This LCD has two registers, command and data. The command register stores the command instructions given to the LCD. A command is an instruction given to the LCD to do a predefined task like initializing it, clearing its screen, setting the cursor position, controlling display etc. The data register stores the data to be displayed on the LCD. The data is the ASCII value of the character to be displayed on the LCD.



Fig. 2.7 LCD Pin Diagram

2.11 BATTERY / POWER SUPPLY

The Automatic Wheelchair runs on a rechargeable 12V or 24V battery. This battery keeps the motors, ESP32 controllers, IR sensors, and the GSM module going. There's a regulated power system inside that makes sure every part gets the right voltage, so things stay steady. With this setup, the battery lasts a long time, so you don't have to keep stopping to recharge. Good power distribution keeps the motors working reliably and the sensors monitoring nonstop.

2.12 BUZZER / LED INDICATORS

The buzzer and LEDs make the Automatic Wheelchair much easier to use. When something's in the way or there's an emergency, the buzzer sounds off right away so the user knows what's happening. The LEDs light up to show things like when the power's on, if there's an obstacle nearby, or if the wheelchair's communicating with another device. With these alerts, users get instant feedback about what's going on, making everything feel safer and more straightforward. Electronic buzzer and its symbol are shown in the Fig. 2.13 (a) and (b) respective.



Fig.2.8 (a) Electronic Buzzer (b) Symbol

2.13 POWER SWITCHES & CONNECTORS

Power switches and connectors keep everything running smoothly inside the Automatic Wheelchair. The main power switch lets users turn the whole system on or off without any hassle, and there are usually extra switches for picking modes or hitting the emergency cutoff if something goes wrong. Connectors and wiring terminals keep the battery, ESP32 controllers, motor drivers, sensors, and all the other pieces linked up in an organized way. This setup delivers steady power and makes it a lot easier to fix problems or handle routine maintenance.

CHAPTER THREE

EXECUTION OF THE PROJECT HARDWARE

3.1 IMPLEMENTATION OF THE HARDWARE

Building the Automatic Wheelchair starts with pulling together all the main parts: ESP32 controllers, a joystick, IR sensors, motor drivers, a GSM module, and DC motors. The joystick and Wi-Fi ESP32 module handle user input and link up with the main ESP32, which runs the show as the central controller.

IR sensors sit up front, scanning for obstacles and sending alerts straight to the ESP32 to keep things safe. Each wheel gets its own DC geared motor, driven by an L298N motor driver—so yeah, the controller can move every wheel on its own. There's a GSM module plugged in through UART, ready to fire off emergency alerts if needed. Power comes from a 12V or 24V battery, which feeds the motors and everything else. The ESP32 and sensors get their own regulated 5V or 3.3V supply. Everything gets mounted firmly onto the wheelchair frame to keep things stable and running smoothly.

3.2 CIRCUIT DIAGRAM OF THE HARDWARE

Take a look at Fig. 3.1 (a) and (b) for the Automatic Wheelchair's circuit diagram. At the heart of everything, you've got the ESP32 microcontroller running the show. Every input, output, and power line hooks up just like you see in the diagram. For controls, the joystick and Wi-Fi ESP32 module both feed into the main ESP32. IR sensors handle obstacle detection. Four L298N motor drivers connect to four DC geared motors to move the wheelchair around. When it comes to emergencies, the GSM module hooks up through the UART pins to send alerts. Power comes from a 12V or 24V battery, and there's a voltage regulator to make sure the ESP32 and sensors get exactly what they need.

Automatic Smart Health-Assisted Wheelchair System

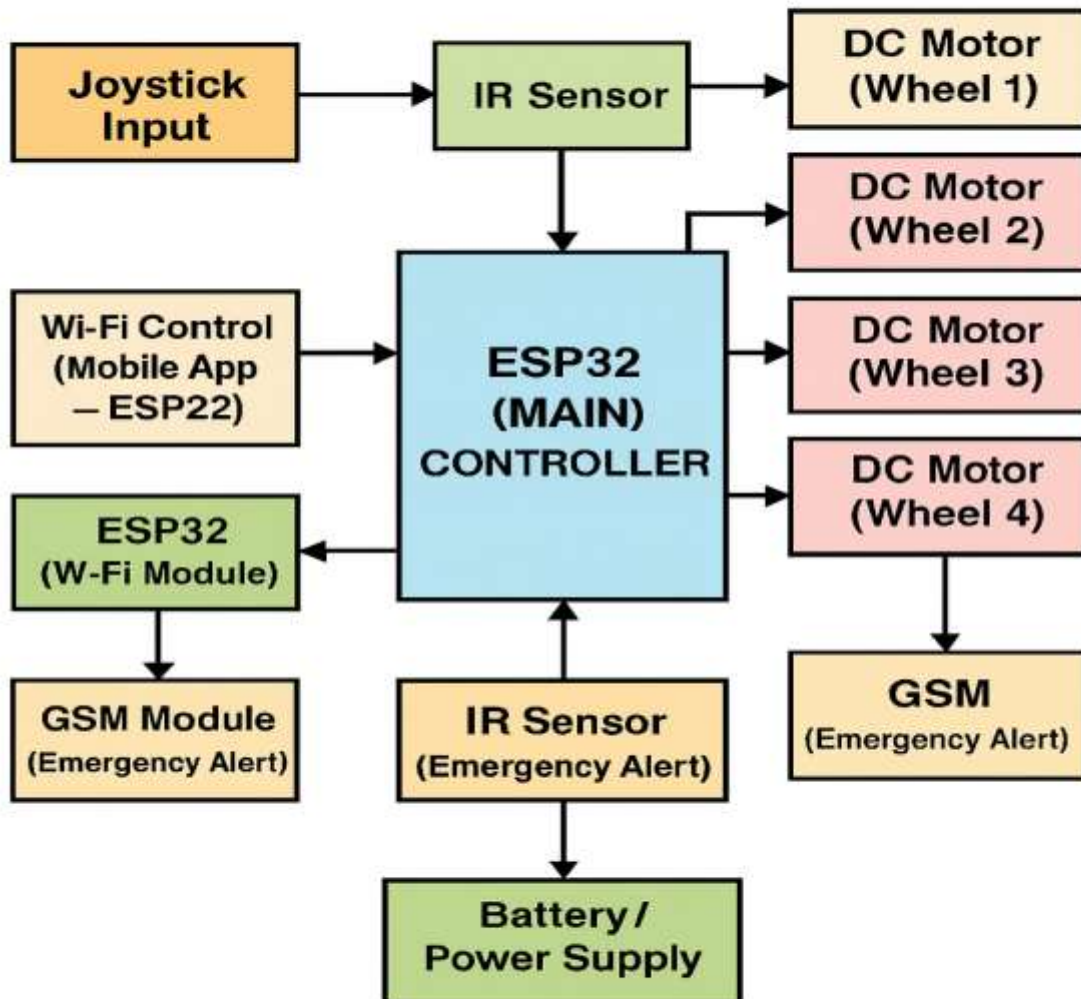


Fig. 3.1 (a) Circuit Diagram of Automatic Wheelchair

3.3 WORKING OF THE PROJECT

The Automatic Wheelchair takes commands from either a joystick or a Wi-Fi app on your phone. The ESP32 microcontroller handles those commands and sends them straight to the four DC motors through L298N motor drivers. IR sensors keep an eye out for obstacles, alerting the controller right away to keep things safe. If there's an emergency, the GSM module steps in to send out alerts. Everything runs on a 12V or 24V battery, so the wheelchair keeps moving smoothly and reliably.

The working of this project is as explained below:

- Push the joystick in any direction—forward, backward, left, or right—and the analog values shift. The ESP32 picks up these changes straight through its ADC pins.
- If you control it from the Wi-Fi app on your phone, the second ESP32 catches those wireless signals and passes them on to the main controller.
- The main ESP32 keeps an eye out for obstacles using IR sensors. When something blocks the path, the IR sensor flips HIGH. That's the warning sign.
- At that point, the controller reacts fast. It either slows down or stops the motors to sidestep a crash. You'll hear a buzzer and see the LED light up—both letting you know there's something in the way.
- Once the obstacle's gone, the IR sensor drops back to LOW. The controller gets things moving again, and both the buzzer and LED switch off on their own.
- The ESP32 sends the right signals to all four L298N motor drivers. Each driver handles one DC geared motor, so the wheelchair moves smoothly.
- If you want to tweak how sensitive the obstacle detection is, or adjust the speed or direction, just update the ESP32's program. It's easy to customize.
- Since the ESP32 can't handle the high current the DC motors need, the L298N motor drivers step in as the bridge between the controller and the motors.
- Press the emergency switch or if something unsafe happens, the GSM module jumps into action and sends emergency SMS alerts.
- A 12V or 24V battery powers the whole setup. The ESP32, sensors, and communication modules get their own regulated 5V or 3.3V supply.

In this chapter we have explained the circuitry, hardware used and working of the proposed project.

CHAPTER FOUR

EXECUTION OF THE PROJECT SOFTWARE

4.1 SOFTWARE FOR PROGRAMMING THE CONTROLLER

We use the Arduino IDE to program the Automatic Wheelchair. It works with the ESP32 boards and makes writing and uploading code pretty straightforward. With Arduino, we can write programs that read the joystick, pick up Wi-Fi commands, process IR sensor data, control the motors, and manage GSM communication—all in one place. Before we start coding, we install the libraries for Wi-Fi, motor control, and communication. Once the program's ready, we compile it and upload the code to both ESP32 boards using a USB-to-UART cable.

4.2 DOWNLOADING ARDUINO IDE (ARDUINO PROGRAMMING SOFTWARE)

We can load new programs on the ESP32 controller using a USB-to-Serial converter through the Arduino IDE. The Arduino IDE is an open-source software and does not require any license for use. It supports Windows, macOS, and Linux operating systems. We can visit the link given

Link:

<https://www.arduino.cc/en/software>

After downloading, we extract and save the Arduino IDE at any location on our computer. To start creating a project, we double-click on the arduino.exe icon as shown in Fig. 4.2. This will open the Arduino IDE window. Next, we configure the ESP32 board for the first time. For selecting the board, we go to:

To install the ESP32 board in your Arduino IDE, follow these next instructions:

In your Arduino IDE, go to **File> Preferences**

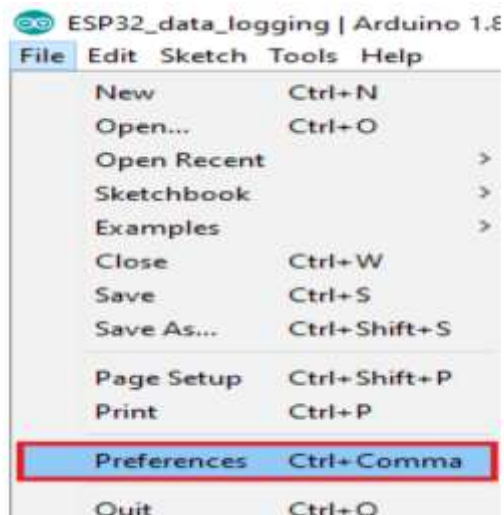


Fig. 4.1 Fig. 4.1 Selection of Preferences

1. Enter the following into the “Additional Board Manager URLs” field:

https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json

Then, click the “OK” button:

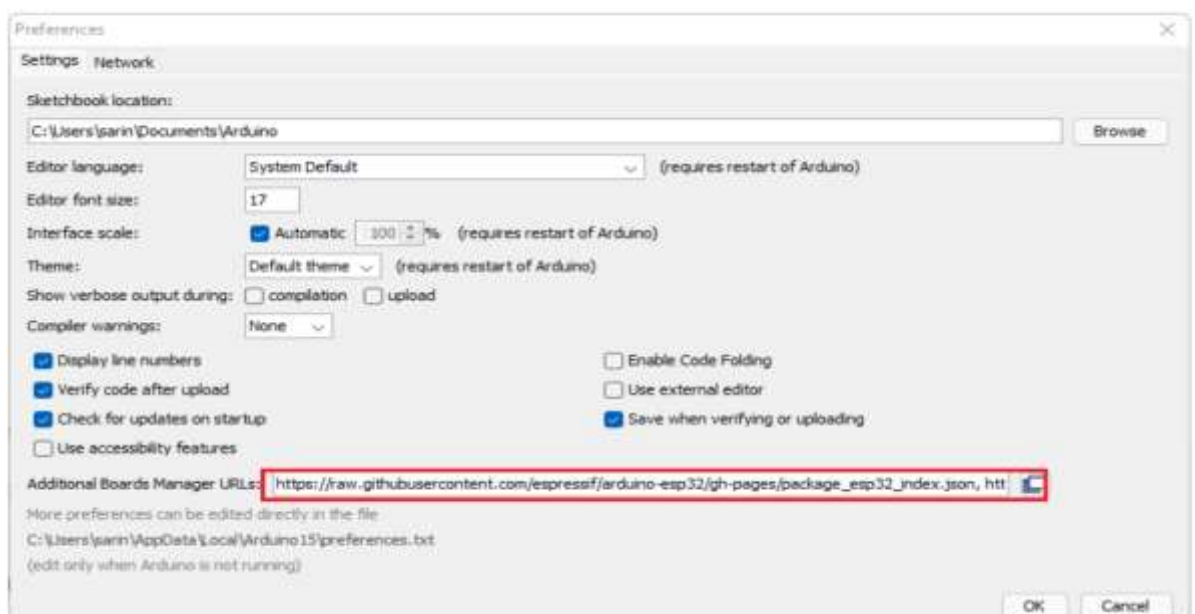


Fig. 4.2 Downloading Additional Board Manager

2. Open the Boards Manager. Go to Tools > Board > Boards Manager

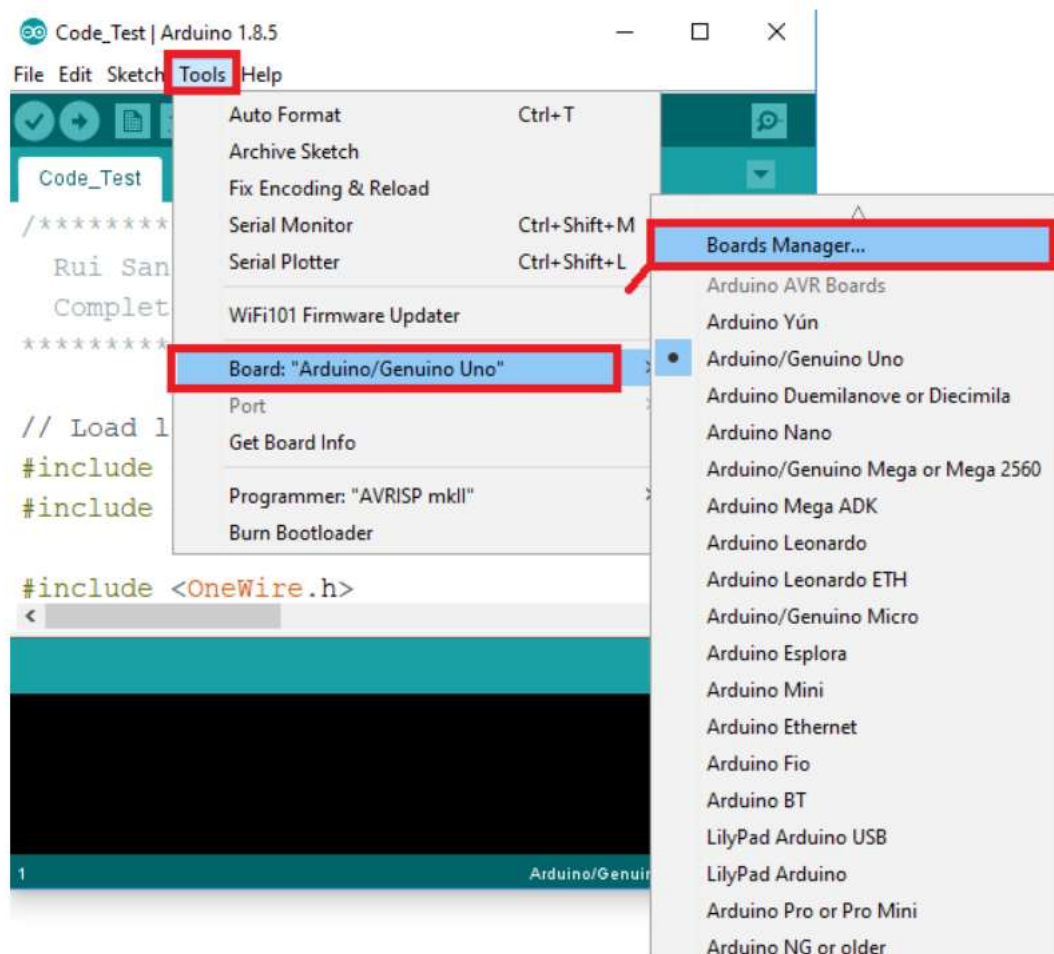


Fig. 4.3 Selection of Board

3. Search for **ESP32** and press install button for the “ESP32 by Espressif Systems”.



Fig. 4.3 Installing ESP 32 Board

4. That's it. It should be installed after a few seconds.

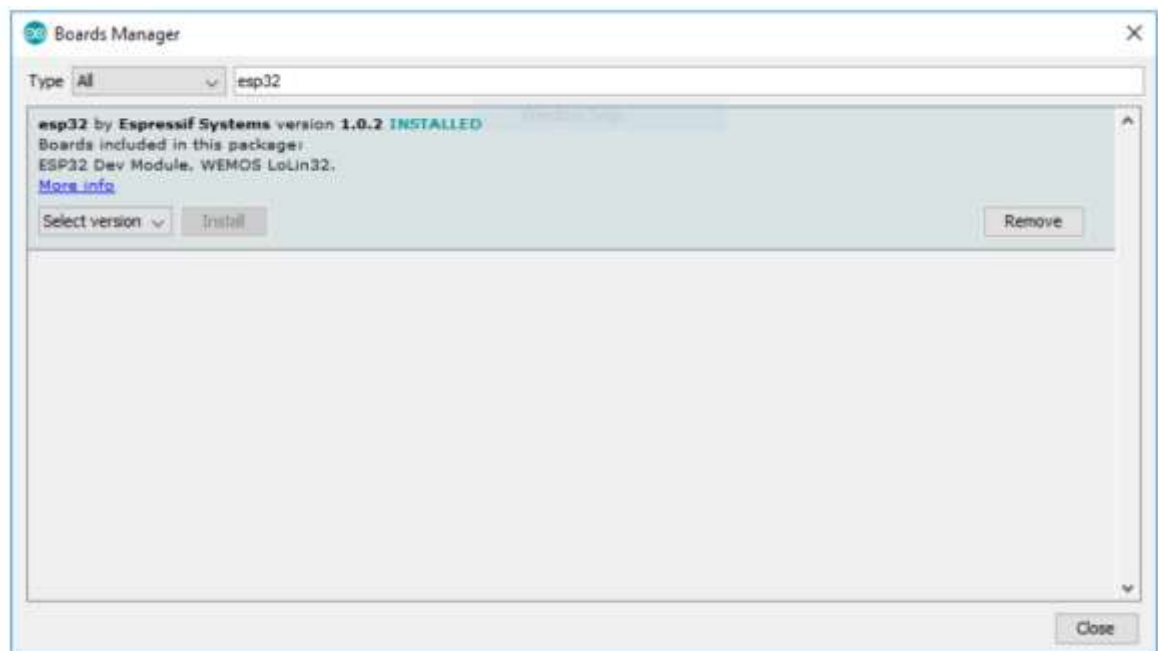


Fig. 4.3 Install Espressif System for ESP 32

5. Plug the ESP32 board to your computer. With your Arduino IDE open, follow these steps:
 1. Select your Board in **Tools > Board** menu (in my case it's the **DOIT ESP32 DEVKIT V1**)

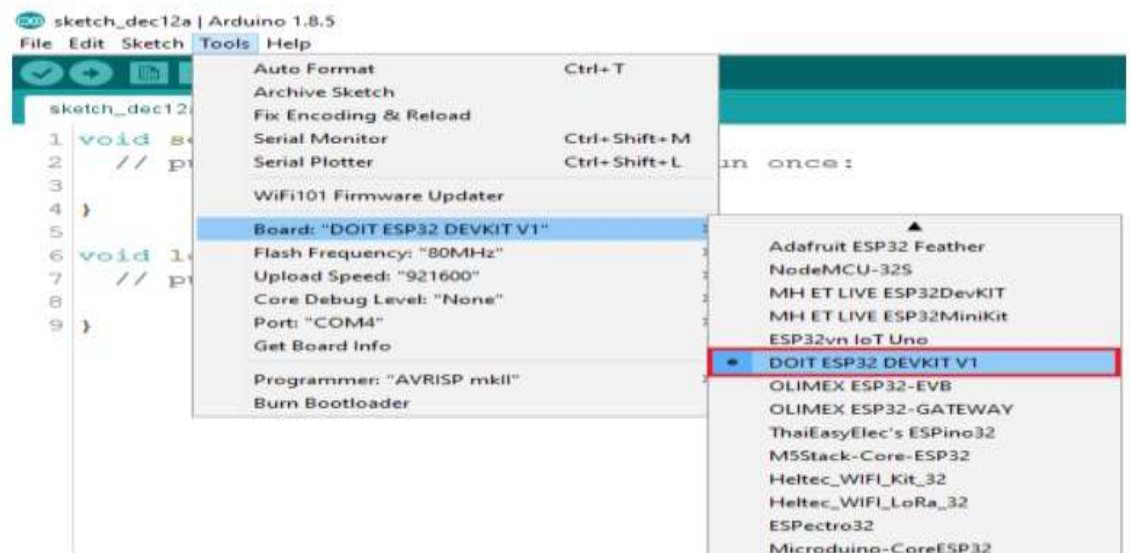


Fig. 4.3 Selection of ESP 32 DEV Board

2. Select the Port (if you don't see the COM Port in your Arduino IDE, you need to install the CP210x USB to UART Bridge VCP Drivers):

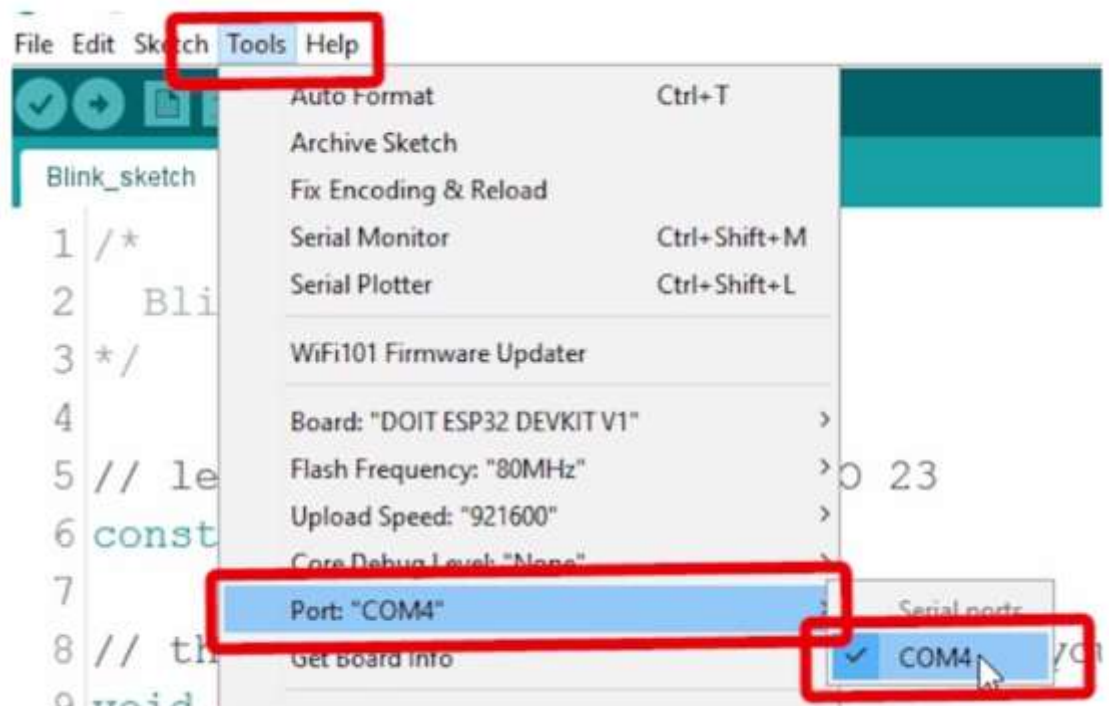


Fig. 4.3 Selection of PORT

In this chapter we have discussed the execution of the software in detail, including the programming process, board configuration, and uploading procedures for the ESP32 controllers. In the next chapter, we are going to explain the results, discussion, and analysis related to the Automatic Wheelchair system.

4.2 Uploading an ESP sketch (Program)

Open the following example under

File > Examples > WiFi (ESP32) > WiFiScan

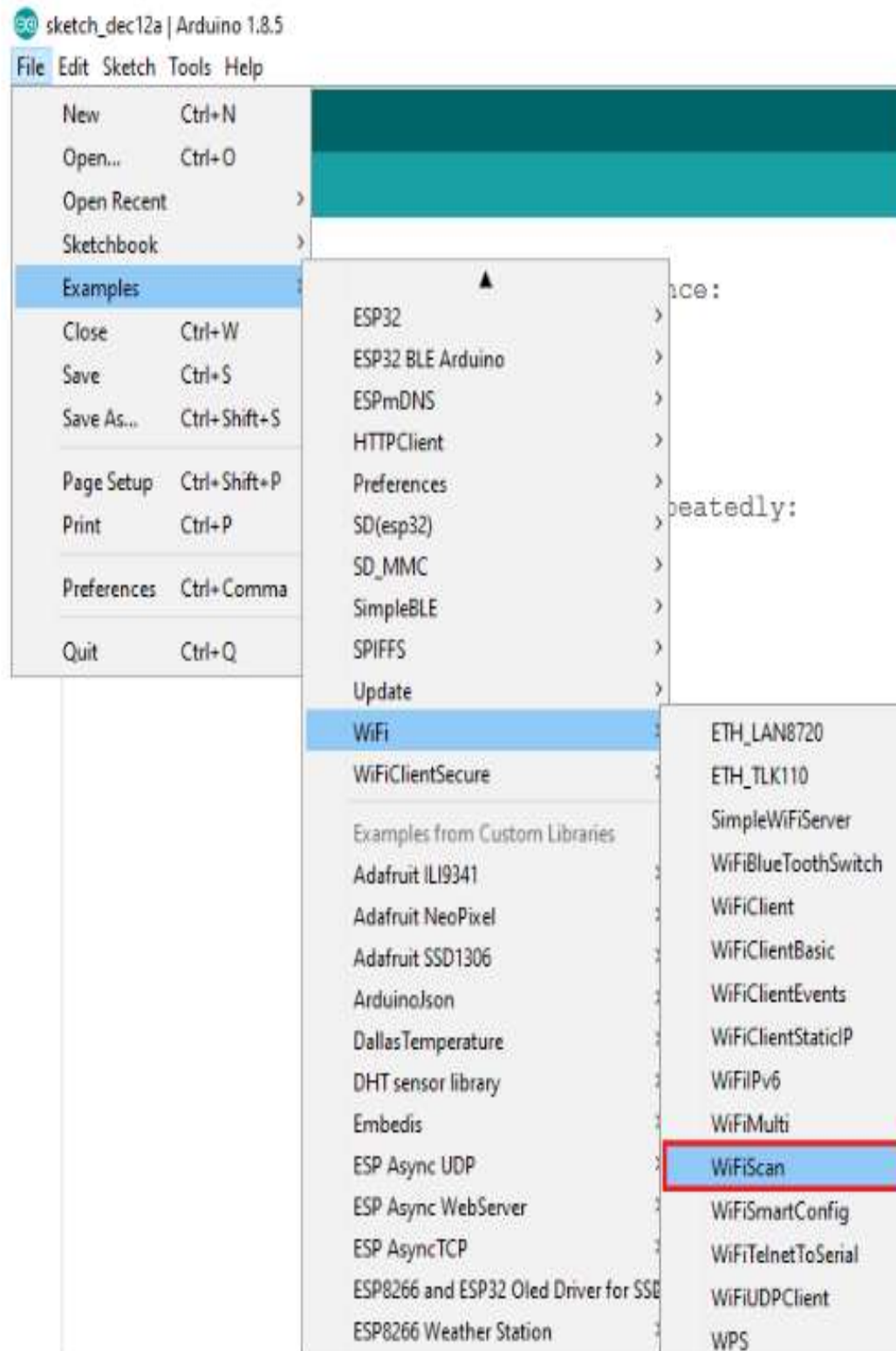
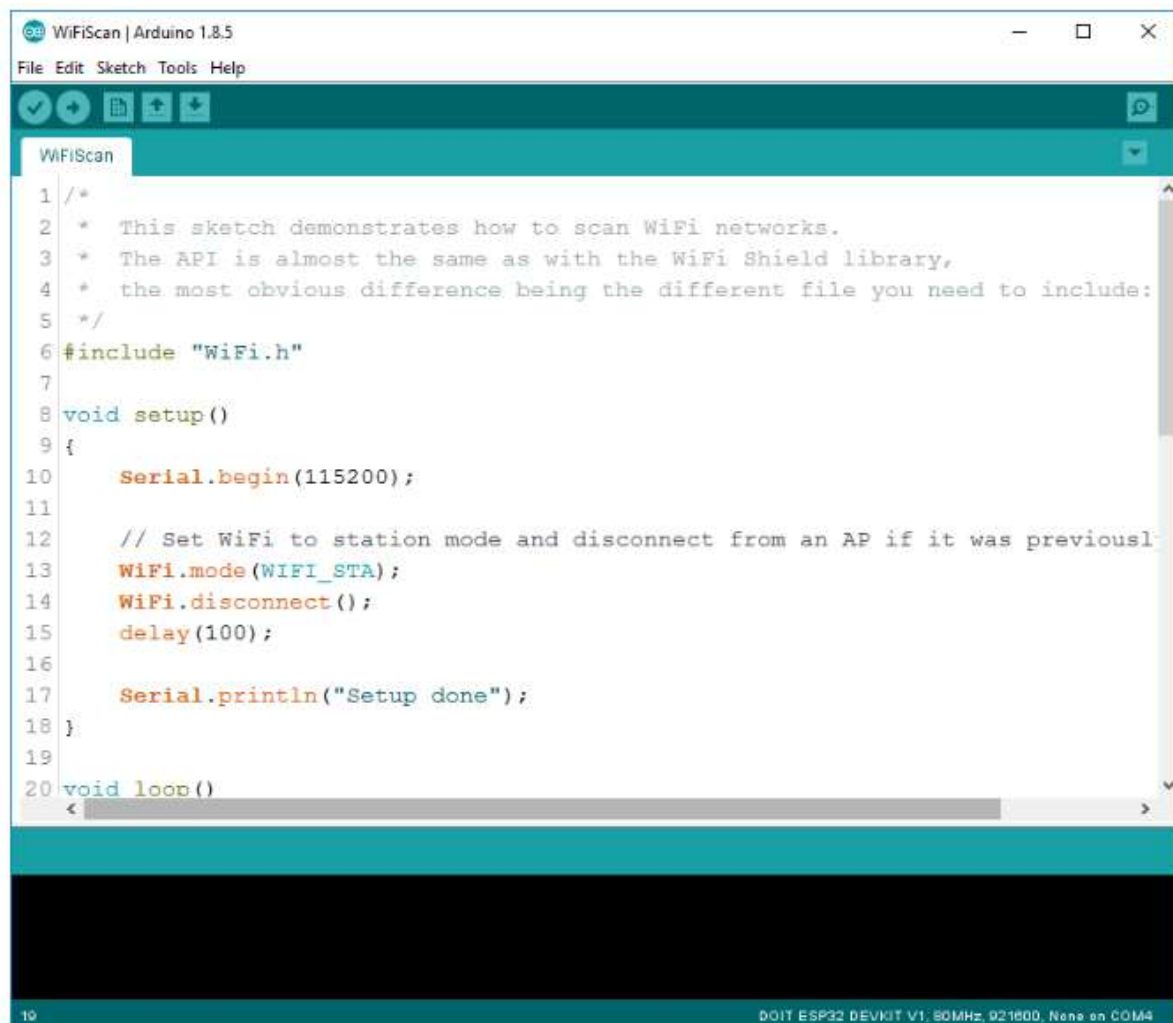


Fig.4.4 Selection of WIFI Scan

A new sketch opens in your Arduino IDE:



```
WiFiScan | Arduino 1.8.5
File Edit Sketch Tools Help

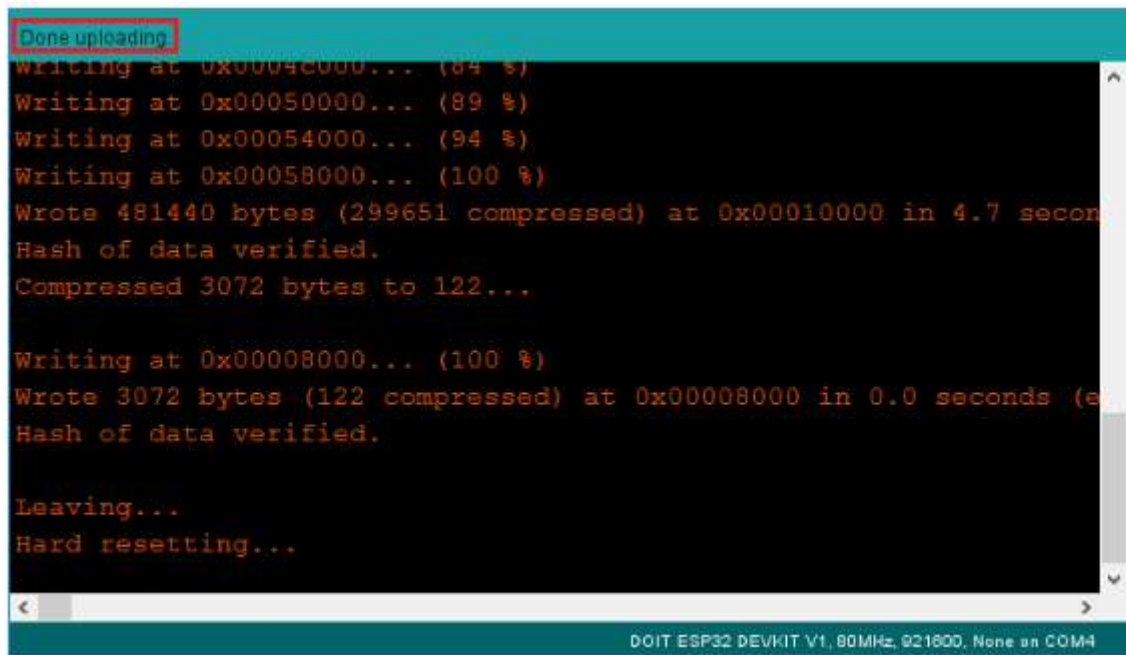
1 /*
2  * This sketch demonstrates how to scan WiFi networks.
3  * The API is almost the same as with the WiFi Shield library,
4  * the most obvious difference being the different file you need to include:
5  */
6 #include "WiFi.h"
7
8 void setup()
9 {
10     Serial.begin(115200);
11
12     // Set WiFi to station mode and disconnect from an AP if it was previousl
13     WiFi.mode(WIFI_STA);
14     WiFi.disconnect();
15     delay(100);
16
17     Serial.println("Setup done");
18 }
19
20 void loop()
```

10 DOIT ESP32 DEVKIT V1, 80MHz, 921600, None on COM4

Fig.4.5 Program of WIFI Scan

Press the **Upload** button in the Arduino IDE. Wait a few seconds while the code compiles and uploads to your board.

If everything went as expected, you should see a “**Done uploading.**” message.



```
Done uploading.
Writing at 0x00042000... (84 %)
Writing at 0x00050000... (89 %)
Writing at 0x00054000... (94 %)
Writing at 0x00058000... (100 %)
Wrote 481440 bytes (299651 compressed) at 0x00010000 in 4.7 seconds
Hash of data verified.
Compressed 3072 bytes to 122...

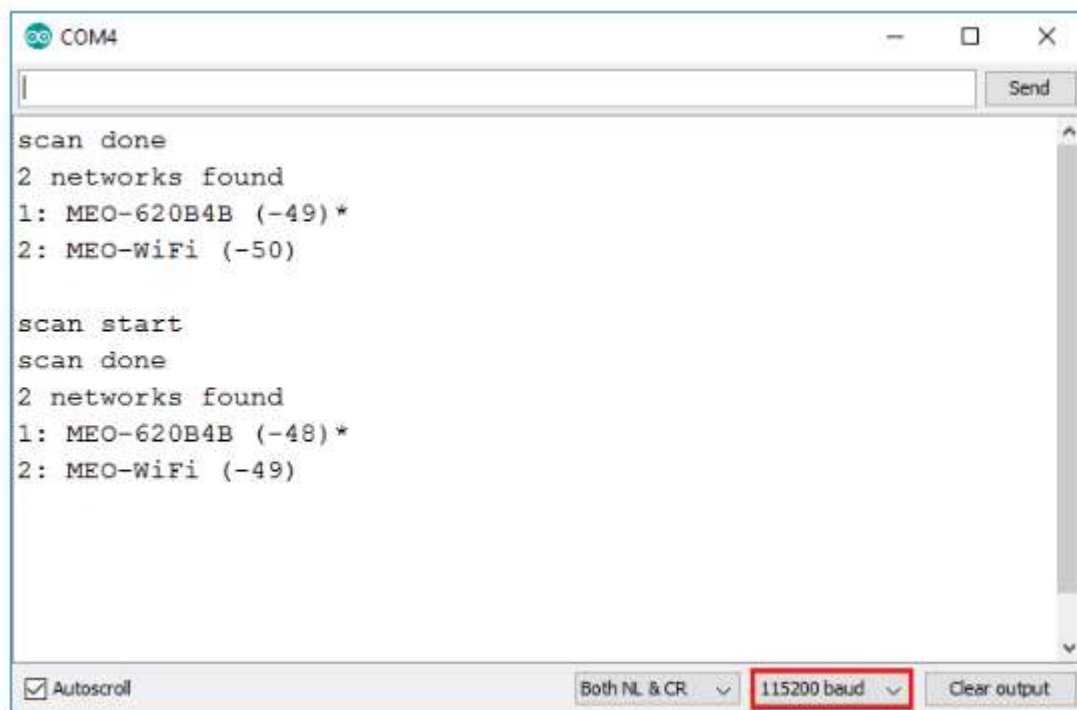
Writing at 0x00008000... (100 %)
Wrote 3072 bytes (122 compressed) at 0x00008000 in 0.0 seconds (effective 115200 bps)
Hash of data verified.

Leaving...
Hard resetting...
```

DOIT ESP32 DEVKIT V1, 80MHz, 921600, None on COM4

Fig.4.6 Uploading Program

Press the ESP32 on-board **Enable** button and you should see the networks available near your ESP32:



```
COM4
scan done
2 networks found
1: MEO-620B4B (-49)*
2: MEO-WiFi (-50)

scan start
scan done
2 networks found
1: MEO-620B4B (-48)*
2: MEO-WiFi (-49)
```

☒ Autoscroll Both NL & CR 115200 baud Clear output

Fig 4.7 Output on Terminal

Troubleshooting

If you try to upload a new sketch to your ESP32 and you get this error message “*A fatal error occurred: Failed to connect to ESP32: Timed out... Connecting...*“. It means that your ESP32 is not in flashing/uploading mode.

Having the right board name and COM port selected, follow these steps:

- Hold-down the “**BOOT**” button in your ESP32 board

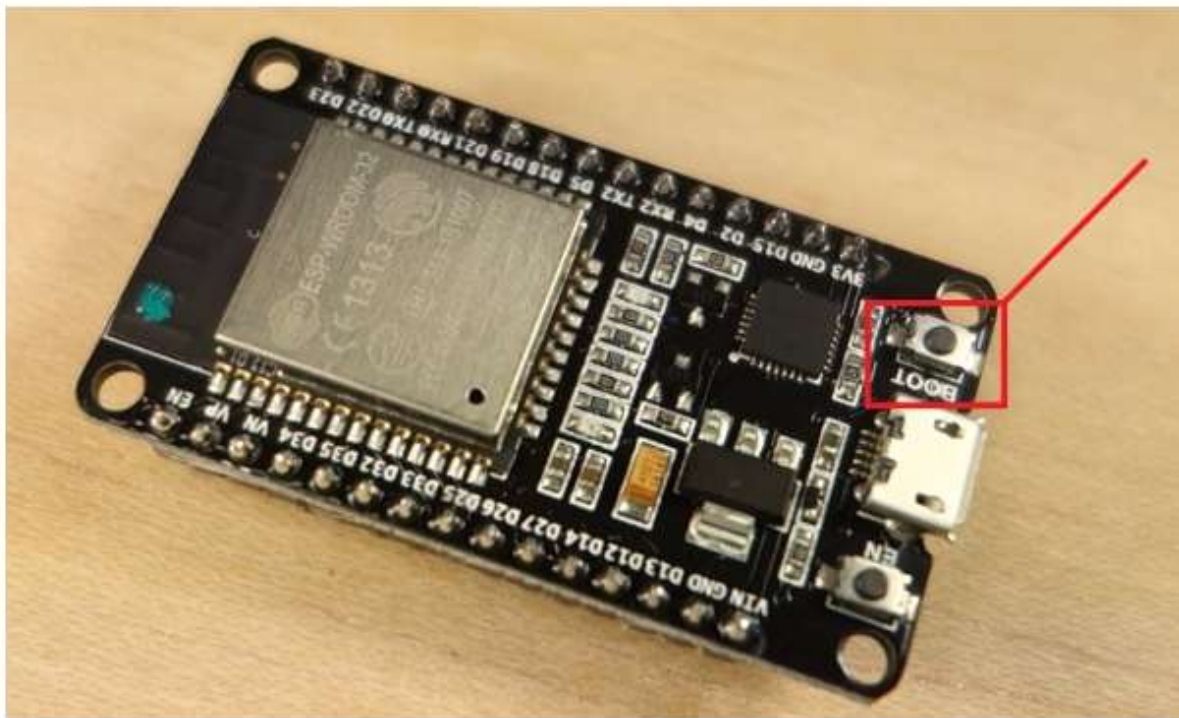


Fig.4.8 Diagram of ESP 32

- Press the “**Upload**” button in the Arduino IDE to upload your sketch:
- After you see the “**Connecting....**” message in your Arduino IDE, release the finger from the “**BOOT**” button:

```
Uploading...
Archiving built core (caching) in: C:\Users\BUTGAR\AppData\Local\Temp\arduino_cache_83883\core\core_espressif_esp32_esp32dev-1.0.0-avr_1.0.0
Sketch uses 501366 bytes (38%) of program storage space. Maximum is 1310720 bytes.
Global variables use 37320 bytes (12%) of dynamic memory, leaving 257592 bytes for local variables. Maximum is 294912 bytes.
upload.py v2.1
Connecting.....
Chip is ESP32DOVE (revision unknown 0x00)
Uploading stub...
Running stub...
Stub running...
Changing baud rate to 921600
Changed.
Configuring flash size...
Auto-detected flash size: 4MB
Compressed 8192 bytes to 47...

Writing at 0x00000000... (100 %)
Wrote 8192 bytes (47 compressed) at 0x00000000 in 3.0 seconds (effective 8192.1 kbit/s)...
Hash of data verified.
Compressed 12304 bytes to 8128...

Writing at 0x00001000... (100 %)
```

Fig 4.9 Error on Terminal

- After that, you should see the “**Done uploading**” message
- After that, you should see the “**Done uploading**” message

CHAPTER FIVE

RESULTS, DISCUSSION AND ANALYSIS

5.1 INTRODUCTION

We built and tested the Automatic Wheelchair to see how it handles different situations. From the start, we focused on keeping things simple, safe, and reliable. For hardware, we went with parts like ESP32 controllers, IR sensors, motor drivers, and DC geared motors—nothing fancy, just stuff that's affordable and easy to find. When we ran our tests, the wheelchair worked smoothly and responded in real time. No lag. No hiccups. Just solid performance.

5.2 RESULTS AND DISCUSSION

The wheelchair responded to joystick commands right away—moving smoothly in every direction without any hiccups. The Wi-Fi app made it simple to control things wirelessly, as long as we stuck to the network range. IR sensors caught obstacles instantly. The second something blocked the way, the motors stopped, and both the buzzer and LEDs kicked in to warn the user.

Each DC geared motor had its own L298N driver, so movements stayed steady and turns felt sharp. In emergency tests, the GSM module sent out SMS alerts to the registered phone number right on time. Everything worked together seamlessly, proving the whole system actually works.

We noticed the power supply stayed stable the entire time, and the battery backup lasted long enough for continuous use. The ESP32 handled inputs fast—there was no lag—so the wheelchair responded in real time. It stayed balanced, even on uneven ground. The wiring and components held up during movement, which shows the hardware assembly is solid. All in all, the system ran smoothly and all features did exactly what they were supposed to during real-world tests.

A hardware model of the system has been developed as shown in Fig. 5.1.



Fig.5.1 Image of Final Project

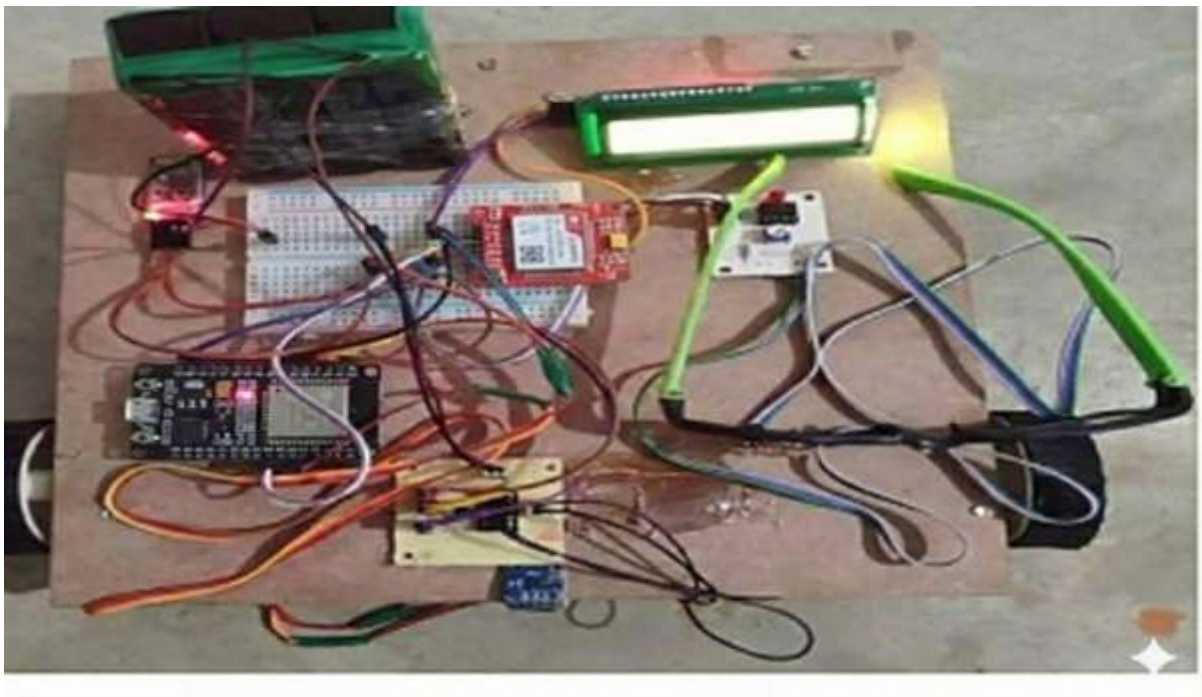
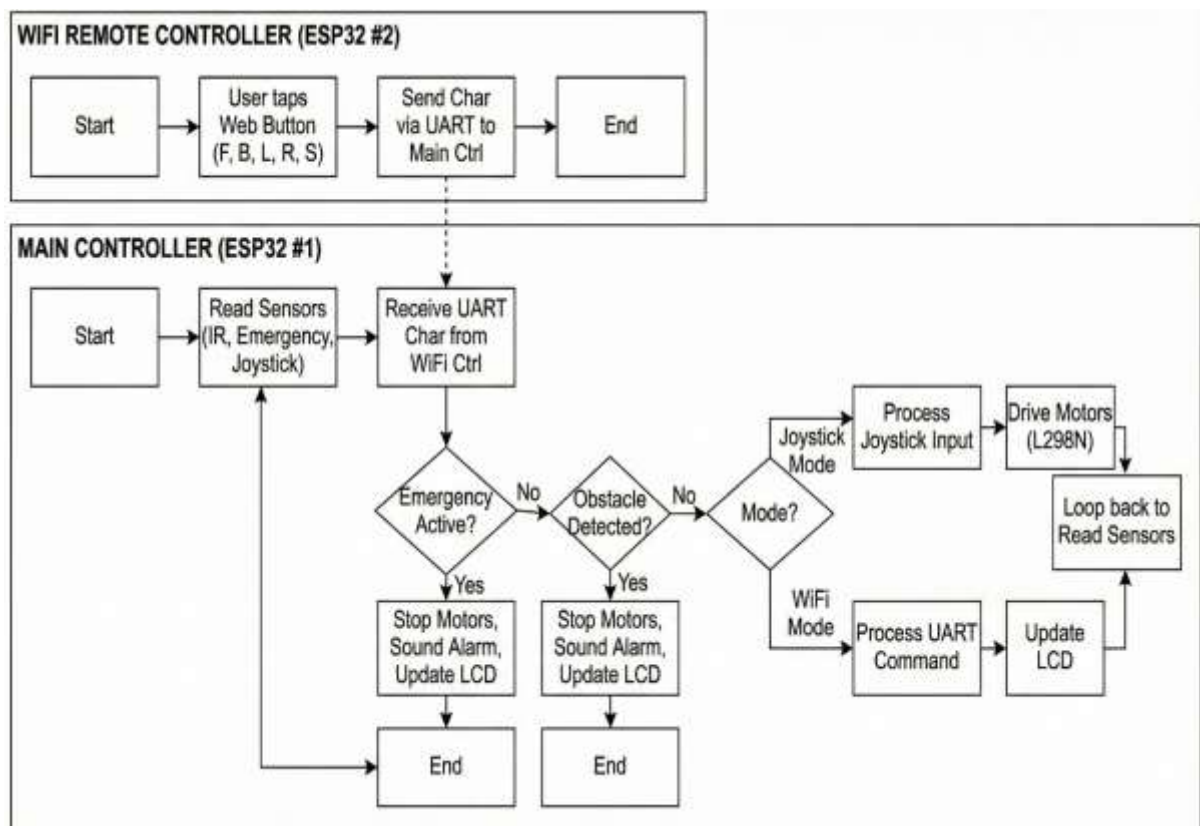


Fig.5.2 Image of Circuit Connection of Project

5.3 FLOW CHART

The system keeps an eye on joystick commands or Wi-Fi input and uses those to steer the wheelchair motors through motor drivers. If an obstacle in the way, the IR sensor spots it and helps prevent any collisions. There's also an emergency button—press it, and the system instantly sends a GSM-based SMS alert to caregivers. You can check the operational mode and status on the 16×2 LCD screen. The control loop keeps everything running smoothly, making sure the wheelchair stays safe and responsive the entire time.

FLOW CHART OF AUTOMATIC WHEELCHAIR



5.4 COST CALCULATION OF THE HARDWARE MODEL

The complete cost calculation of the Automatic Wheelchair hardware is shown in Table 5.1. The analysis indicates that the system is economical and can be implemented at a low cost compared to commercial automatic wheelchairs.

Table 5.1 Cost of the Hardware Model

S. No	Hardware Component	Specification	Quantity	Unit Cost (₹)	Total Cost (₹)
1	ESP 32	Microcontroller for processing and control	2	800	1600
2	DC / Stepper Motors	DC / Stepper Motors	4	700	2800
3	LED Display	Character display with backlight	1	400	400
4	Motor Driver Module	Motor Driver Module	4	500	2000
5	Heart Rate Sensor	MAX30100	1	400	400
6	Temperature Sensor	LM35	1	400	400
7	12V DC Relay	Single channel relay module	4	200	800
8	Joystick Module	Manual control input	1	500	500
9	Buzzer	5V audio buzzer	1	100	100
10	Power Supply	12V rechargeable battery with BMS	1	2000	2000
11	IR Sensor	IR Sensor	1	500	500
12	Chair Body	Chair Body	1	2000	2000
13	Breadboard	Circuit mounting board	2	300	600
14	Miscellaneous	Jumper wires and cables, Jumper wires and cables	As per use	1000	1000
TOTAL					15100 ₹

CHAPTER SIX

CONCLUSIONS AND FUTURE SCOPE

6.1 CONCLUSIONS

This project's Automatic Wheelchair turned out to be a straightforward, reliable, and affordable way to help people move around more easily. It blends joystick and Wi-Fi controls, so users get smooth, flexible handling. During testing, the IR sensors picked up obstacles every time, stopping the chair before any bumps could happen. Thanks to the four motors and four drivers, movement felt steady and turning was easy.

The GSM emergency alert was a real plus. It gave users a fast way to reach out if something went wrong. All in all, the wheelchair handled every command without a hitch, worked well in different situations, and just made sense for everyday use. In the end, it's a solid automatic mobility aid that actually delivers.

6.2 FUTURE SCOPE

The proposed hardware can be improved and expanded in the future with the following possible enhancements:

- You can add a speed-control feature so users set the wheelchair's pace just how they like it. Throw in some advanced sensors—ultrasonic or LiDAR—and you'll get much better obstacle detection, both in range and accuracy. GPS tracking is another solid upgrade, letting you handle outdoor navigation and check real-time location whenever you need.
- A mobile app opens up a bunch of new options—battery status, system diagnostics, all right at your fingertips. For folks with severe physical limitations, voice commands or gesture controls make things a lot more accessible. Cameras can help too, giving the wheelchair smart navigation, object detection, and extra help in tricky spaces.
- Hooking the system into an IoT platform means caregivers can keep an eye on things remotely and jump in when needed. Swap in rechargeable lithium-ion battery packs, and you get longer operating times and fewer charging breaks. A solar charging panel helps keep the battery topped up while you're outdoors.

- Using brushless DC motors (BLDC) bumps up efficiency, gives you smoother speed control, and they just last longer. Load sensors watch how the user's weight shifts—helpful for avoiding accidents, especially on slopes. A smart braking system smooths out stops and keeps things safe.
- Bluetooth, along with Wi-Fi, gives you more ways to control the chair wirelessly. Building a foldable frame makes it easier to take the wheelchair with you, anywhere. With autonomous indoor mapping, users who can't steer much on their own still get around just fine.
- Upgrade the suspension, and you'll notice the ride is a lot more comfortable over bumps and rough ground. Add a health monitoring module—pulse, temperature, SpO₂—and you can track health in real time. Connecting to the cloud lets you store data and monitor the chair from anywhere.
- A mobile dashboard for caregivers shows alerts, location, and battery info all in one spot. A reverse movement sensor helps prevent backing into things. Waterproofing and shock proofing the electronics keeps everything safe outdoors. And for hands-free help, why not include AI-based voice assistants like Google Assistant or Alexa?

REFERENCES

1. M. R. Yuce and J. Khan, *Wireless Body Area Networks: Technology, Implementation and Applications*, Pan Stanford Publishing, ISBN 978-981-431-6712, 2011.
2. M. M. Khan, "IoT based smart healthcare services for rural unprivileged people in Bangladesh: current situation and challenges," in 1st International Electronic Conference on Applied Science, MDPI, pp. 1–6, Switzerland, 2020.
3. <https://randomnerdtutorials.com/installing-the-esp32-board-in-arduino-ide-windows-instructions/>.
4. S. Gupta, "Design of Wi-Fi Controlled Wheelchair for Disabled People," *IRJET Journal*, Vol. 11, Issue 2, 2024.
5. M. Sharma, "Obstacle Detection Methods Using Infrared Sensors," *Journal of Smart Electronics*, 2023.
6. Arduino IDE Installation and ESP32 Board Setup Guide," Arduino.cc, <https://www.arduino.cc/en/software>.
7. Random Nerd Tutorials, *Installing ESP32 Board in Arduino IDE – Step-by-Step Guide*, Available at: <https://randomnerdtutorials.com/installing-the-esp32-board-in-arduino-ide-windows-instructions/>.
8. Nevon Projects, *GSM Module Interfacing and Applications in IoT*, 2023.
9. STMicroelectronics, *L298N Dual H-Bridge Motor Driver Datasheet*, 2023.
10. IEEE IoT Journal, "Emergency Alert Systems Using GSM Technology," Vol. 9, Issue 4, 2024.
11. S. Gupta, "Design of Wi-Fi Controlled Wheelchair for Disabled People," *IRJET*, Vol. 11, Issue 2, 2024.
12. Electronicsforu.com, "Motor Driver Control for Wheel-Based Robots," EFY Publications, 2023.